B2.3 Cell specialization

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b2-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 265–272 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Exchange surfaces

B2.3.7 Raising the surface area-to-volume ratio

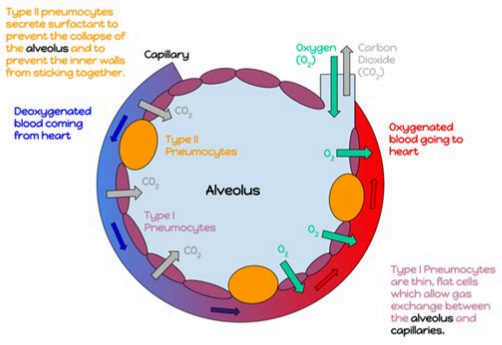
pp. 265–266

Cells increase their surface area-to-volume ratio by **flattening**, by **microvilli** (finger-like projections of the plasma membrane) and by **invaginations** (infoldings). Red blood cells are about 8 µm across, biconcave and have no nucleus, so oxygen diffuses only a short way to any haemoglobin, and the cells bend through capillaries. Cells of the **proximal convoluted tubule** have microvilli facing the filtrate, invaginations facing the blood and many mitochondria, for fast active reabsorption.

B2.3.8 Two kinds of pneumocyte

pp. 266–268

Two cell types line each alveolus — B2.3.8



	Type I	Type II
Shape	Very thin, flat	Cuboidal
Surface share	About 95%	About 5%
Features	Few organelles	Lamellar bodies, RER, Golgi
Role	Gas exchange	Secrete surfactant

Alveolus and its cell types. Katherinebutler1331, CC BY-SA 4.0.

Two jobs
Very thin walls and a supply of surfactant: no single cell type could do both well.

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Type I pneumocytes are extremely thin and cover most of the alveolar surface, so the diffusion distance for gases is very short. **Type II** pneumocytes are cuboidal and full of organelles; their **lamellar bodies** release surfactant by exocytosis. Surfactant lowers the surface tension of the water lining the alveoli, so they do not collapse when we breathe out. The alveolar epithelium is a tissue with more than one cell type, because no single cell could do both jobs.

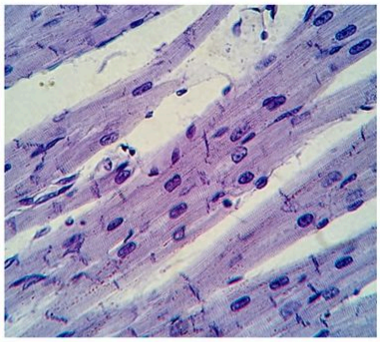
Muscle and gametes

B2.3.9 Cardiac muscle and striated muscle

pp. 268–270

Heart muscle and skeletal muscle

B2.3.9



Cardiac muscle (light micrograph). A. G. Cheroske, CC BY-SA 4.0.



Striated (skeletal) muscle. CC BY-SA 3.0.

Shared
Both striated (sarcomeres of actin and myosin) with many mitochondria.

Look
Find the branches and intercalated discs in heart muscle, and the many nuclei in skeletal fibres.

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Both are striated, because **myofibrils** of actin and myosin are arranged in repeating sarcomeres, and both have many mitochondria. Cardiac muscle cells are short, **branched**, have one nucleus and are joined by intercalated discs, so the signal to contract spreads quickly through the heart. Striated muscle fibres are long, unbranched and have **many nuclei**, because they form by fusion of many cells. A fibre has one plasma membrane and normal organelles, but its size and many nuclei make it an unusual cell.

B2.3.10 Sperm and egg cells

pp. 271–272

A **sperm** is small and streamlined: an acrosome with enzymes to digest the zona pellucida, a haploid nucleus, a mid-piece packed with mitochondria and a flagellum. An **egg** is much larger: its cytoplasm holds nutrients, organelles and mRNA for early development. It is surrounded by the **zona pellucida** and follicle cells, and its **cortical granules** harden the zona after one sperm enters, preventing polyspermy.

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